

Exploratory Data Analysis of Pseudomonas putida Cultivations

Four-variable representation: glucose, dissolved oxygen, pH and biomass

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Dataset	29 AMBR cultivation profiles
Representation	4 process/biological variables
Feature extraction	6 descriptors per variable; 24 initial features
Reduced representation	17 features after correlation filtering
Clustering shown	Exploratory K-means solution with k = 5
Purpose	Describe major cultivation patterns and within-cluster heterogeneity

KEY FINDING

The analysis compresses dynamic cultivation trajectories into interpretable summary descriptors, then uses PCA and clustering to identify recurring cultivation phenotypes.

WHAT IT MEANS - The resulting groups are exploratory patterns in the measured data, not predefined biological classes. They should be validated against cultivation conditions and the original time-series trajectories.

Scope note

This report is based only on the executed four-variable notebook supplied for this analysis. Volume and off-gas CO2 are not part of this representation. Biomass is derived from optical density using the conversion implemented in the notebook.

1. Biological question and analytical strategy

The aim is to compare cultivation runs using the overall shape and level of four biologically meaningful signals: glucose availability, dissolved oxygen (DO), pH and biomass. Because each signal is a time series, direct visual comparison across 29 runs becomes difficult. The notebook therefore converts each trajectory into six descriptors: normalized area under the curve (AUC), maximum temporal slope, maximum, minimum, initial value and final value.

This produces 24 candidate features per cultivation. The features are then filtered for redundancy, standardized, projected with principal component analysis (PCA), and clustered using K-means. A final analysis examines both differences between clusters and variability among experiments within the same cluster.

KEY FINDING

Twenty-nine cultivations are represented by 24 summary features derived from glucose, DO, pH and biomass trajectories.

WHAT IT MEANS - This converts complex time-series behavior into a common quantitative representation that can be compared across reactors while retaining information about levels, endpoints and dynamics.

Analysis workflow

Step	Purpose
Feature extraction	Summarize each trajectory using AUC, slope, max, min, first and last values.
Correlation filtering	Remove strongly redundant descriptors using $ r > 0.85$.
Standardization	Put features with different units on comparable scales.
PCA	Identify dominant multivariate patterns.
K-means	Explore recurring cultivation profiles in PCA space.
Intra-cluster analysis	Measure how homogeneous or heterogeneous each group is.

2. Reducing redundant information

Strongly correlated features can cause the same biological pattern to be counted several times. The notebook therefore removes one feature from each highly correlated pair above an absolute correlation threshold of 0.85. Seven of the 24 initial features were removed: First_Glucose, Last_Glucose, Last_DO, Max_pH, Min_pH, First_pH and Last_pH. Seventeen features remained for PCA and clustering.

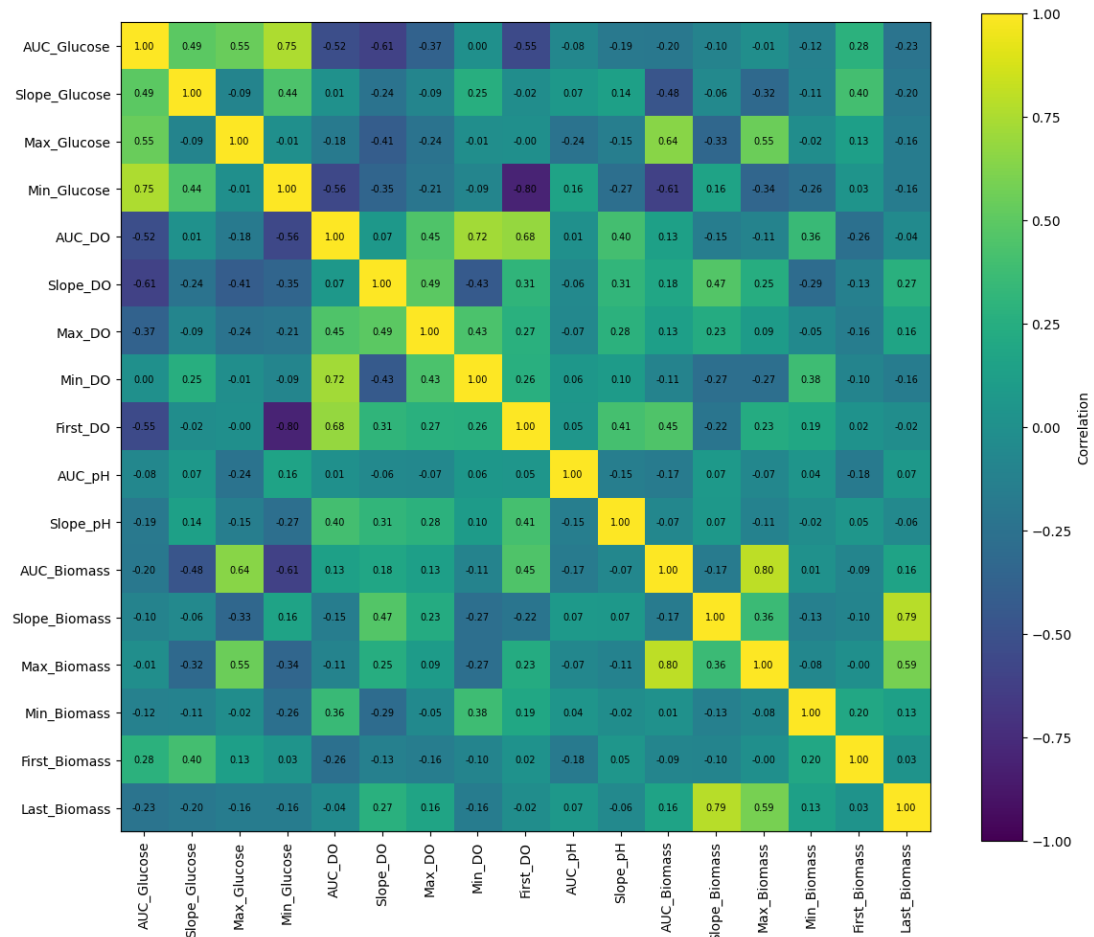


Figure 1. Correlation matrix after redundancy filtering. Values close to +1 or -1 indicate strong linear relationships between retained descriptors.

KEY FINDING

Correlation filtering reduced the representation from 24 to 17 features.

WHAT IT MEANS - The downstream PCA and clustering are less dominated by duplicated information. The remaining features still represent all four biological variables, but not every original descriptor.

3. Principal component analysis: the main axes of variation

PCA rotates the standardized feature space into new axes that summarize combinations of features. PC1 explains 24.8% of the total variance and PC2 explains 17.7%; together they capture 42.5%. The first five components explain approximately 76.3%, which is why the clustering step uses the first five PCs rather than only the two axes shown in the 2D figure.

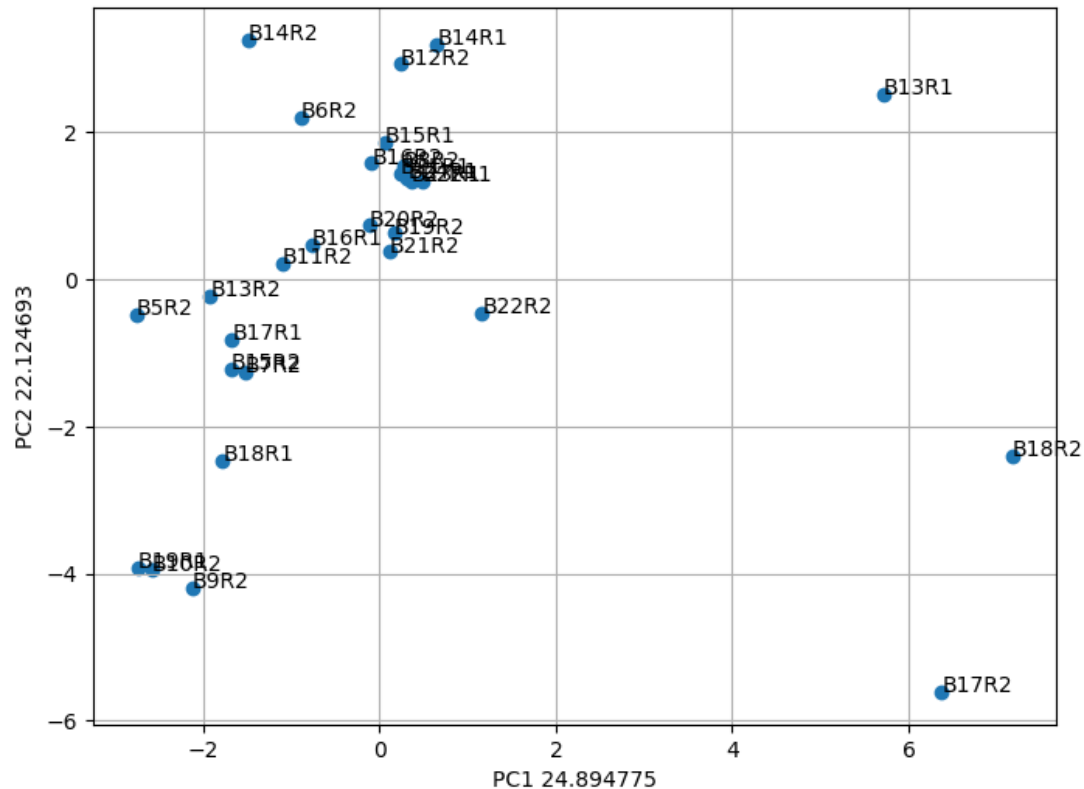


Figure 2. PCA projection of the 29 cultivation profiles on PC1 and PC2. Labels identify bioreactor/run combinations.

KEY FINDING

The first two PCs provide a useful visual overview but contain only 42.5% of the total multivariate variation; five PCs are needed to reach about 76%.

WHAT IT MEANS - Apparent proximity in the 2D plot should not be treated as the complete similarity between cultivations. The clustering uses a richer five-dimensional PCA representation.

4. What drives PC1 and PC2?

PCA loadings indicate which original features contribute most strongly to each axis. PC1 is driven positively by glucose descriptors, especially minimum glucose (loading +0.43) and glucose AUC (+0.40), and negatively by DO descriptors such as initial DO (-0.37) and DO AUC (-0.32). PC2 contrasts DO-related features, particularly minimum DO (+0.43) and DO AUC (+0.37), with biomass accumulation descriptors such as maximum biomass (-0.43), final biomass (-0.32) and biomass slope (-0.31).

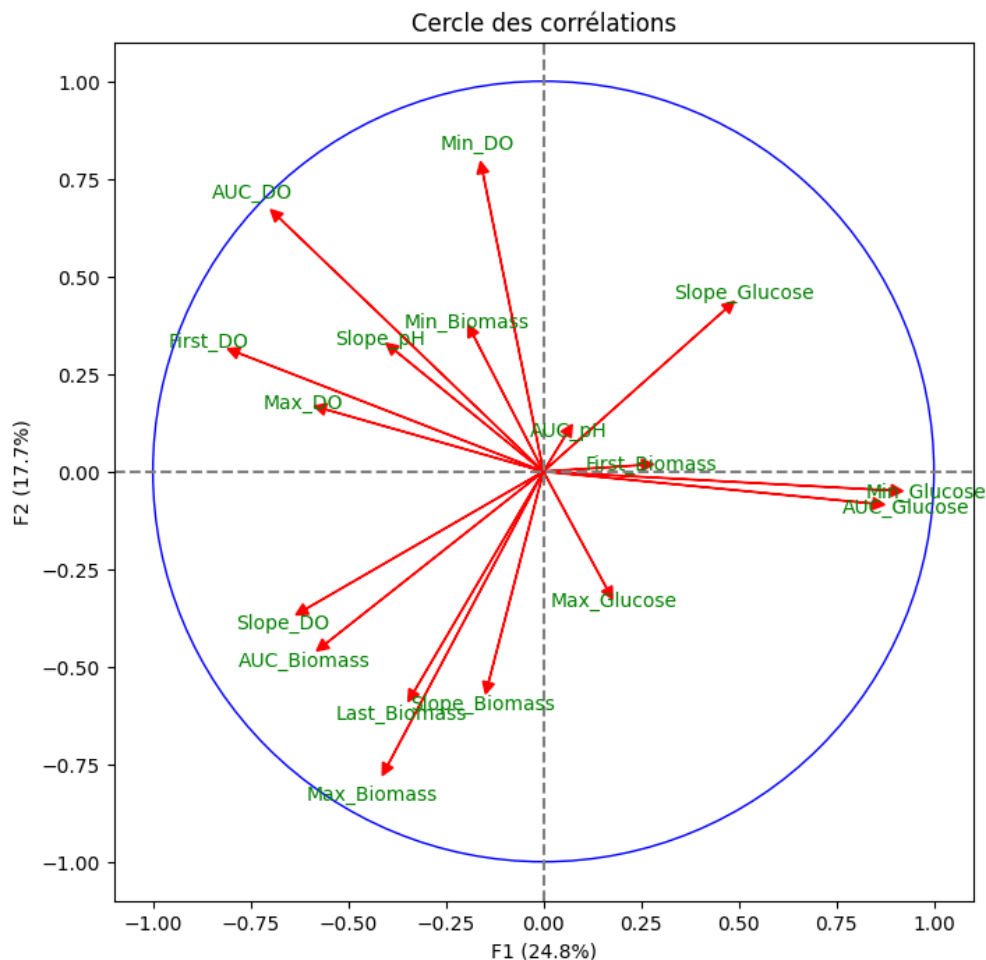


Figure 3. Correlation circle for PC1 and PC2. Features pointing in similar directions vary together; opposing directions indicate inverse multivariate associations.

KEY FINDING

PC1 mainly contrasts glucose-rich profiles with stronger DO/biomass signatures, while PC2 separates DO-related behavior from biomass accumulation patterns.

WHAT IT MEANS - These axes provide biological context for the PCA map, but they describe statistical associations rather than causal relationships. Interpretation should be checked against the original trajectories and experimental conditions.

5. Exploratory clustering of cultivation profiles

K-means was applied to the first five PCs. The notebook retains a five-cluster solution to examine finer substructure among the cultivation profiles. Importantly, this is not the silhouette-optimal number of clusters: $k = 2$ gives the highest silhouette score (0.428), while $k = 5$ gives 0.376. The five-cluster solution should therefore be viewed as an exploratory subdivision rather than evidence for five definitive biological classes.

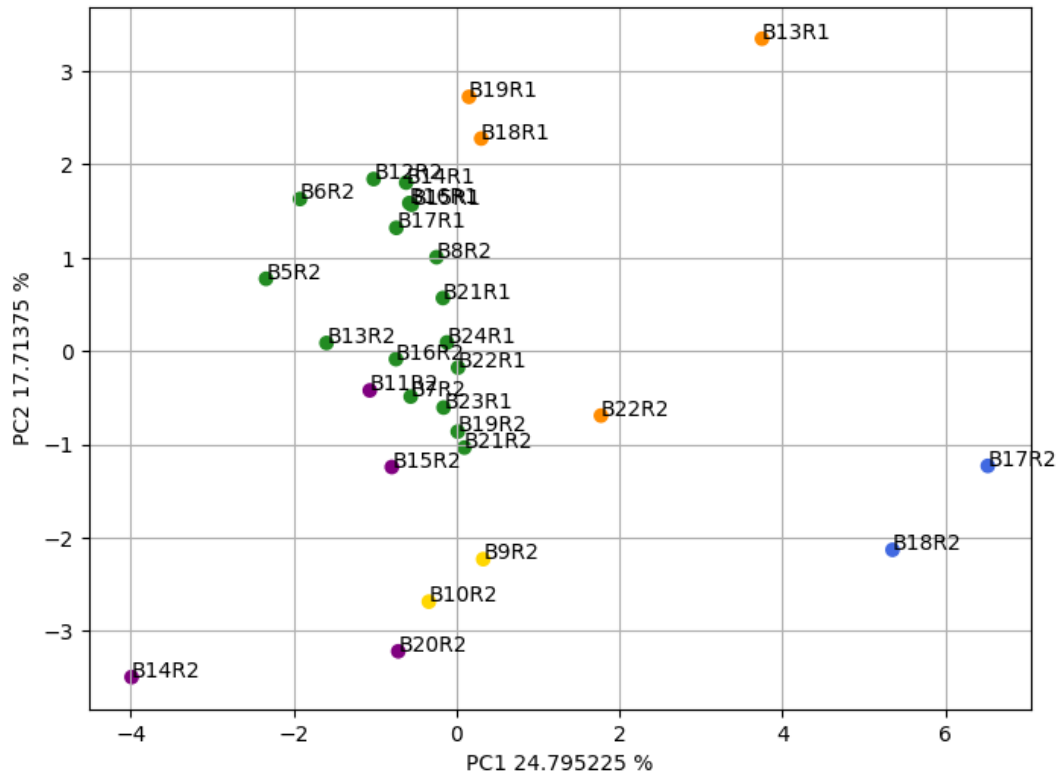


Figure 4. Five-cluster exploratory K-means solution shown on the first two PCA axes. Clustering itself was performed using PCs 1-5.

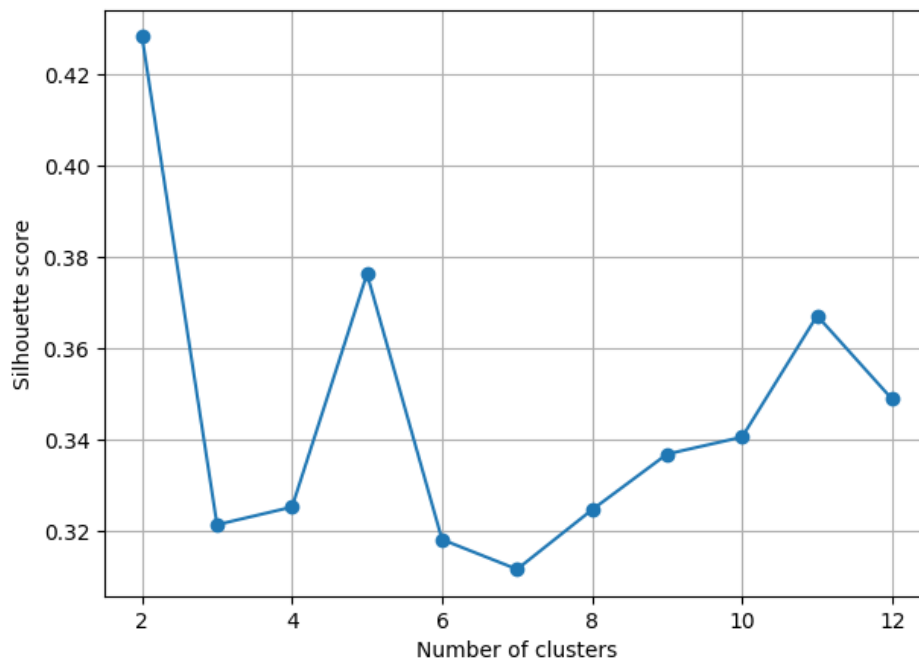


Figure 5. Silhouette score as a function of k . The highest score occurs at $k = 2$; $k = 5$ is retained for finer exploratory interpretation.

KEY FINDING

The strongest global separation is obtained with two clusters, but $k = 5$ provides a finer exploratory partition with a silhouette score of 0.376.

WHAT IT MEANS - The five groups are useful for describing subtypes and unusual profiles, but should not be presented as the uniquely optimal or definitive biological grouping.

6. Biological interpretation of the five exploratory groups

The cluster profiles below are expressed as standardized feature means (z-scores). Positive values indicate that a cluster is above the overall dataset mean for a feature; negative values indicate below-average values. The most discriminating features across clusters are biomass AUC, maximum glucose, initial DO, minimum glucose and maximum biomass.

Cluster	n	Cultivations
0	2	B17R2, B18R2
1	4	B13R1, B18R1, B19R1, B22R2
2	17	B5R2, B6R2, B7R2, B8R2, B12R2, B13R2, B14R1, B15R1, B16R1, B16R2, B17R1, B19R2, B21R1, B21R2, B22R1, B23R1, B24R1
3	4	B11R2, B14R2, B15R2, B20R2
4	2	B9R2, B10R2

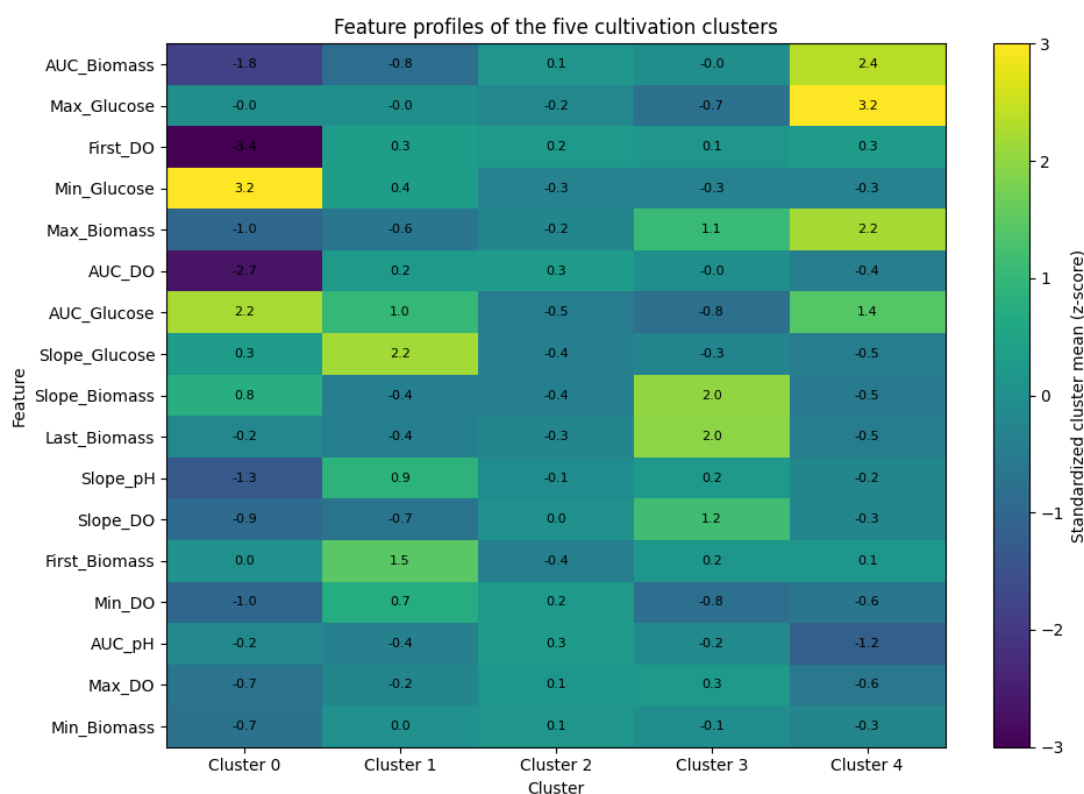


Figure 6. Standardized feature profiles of the five exploratory clusters. Larger absolute z-scores indicate stronger departures from the overall dataset mean.

Cluster 0: B17R2 and B18R2 are strongly distinguished by very low initial DO and DO AUC, while minimum glucose and glucose AUC are high. Biomass AUC is low.

Cluster 1: Four cultivations show particularly high glucose slope and high initial biomass, with lower biomass AUC on average.

Cluster 2: This large central group contains 17 cultivations and is comparatively close to the dataset average. Its strongest deviations are modestly lower glucose AUC, glucose slope, biomass slope and initial biomass.

Cluster 3: Four cultivations show high biomass slope and final biomass, together with elevated DO slope and relatively low minimum DO.

Cluster 4: B9R2 and B10R2 show very high maximum glucose, biomass AUC and maximum biomass, while pH AUC is comparatively low.

KEY FINDING

The five-cluster subdivision separates several distinctive cultivation phenotypes, while most experiments fall into a broad central Cluster 2.

WHAT IT MEANS - The strongest group differences involve glucose availability, oxygen behavior and biomass accumulation. These signatures are hypotheses for biological/process interpretation and should be connected to media, feed, aeration or other experimental metadata before assigning mechanisms.

7. Intra-cluster variability: how similar are members of each group?

Cluster means can hide substantial diversity. To quantify within-cluster heterogeneity, each cultivation was measured by its Euclidean distance from its cluster centroid in the standardized 17-feature space. Smaller distances indicate that a cultivation is more typical of its assigned cluster; larger distances indicate a more unusual member.

Cluster	n	Mean distance	SD	Maximum
0	2	1.53	~0	1.53
1	4	3.92	0.88	4.58
2	17	2.49	0.88	4.73
3	4	3.33	0.76	4.34
4	2	1.09	~0	1.09

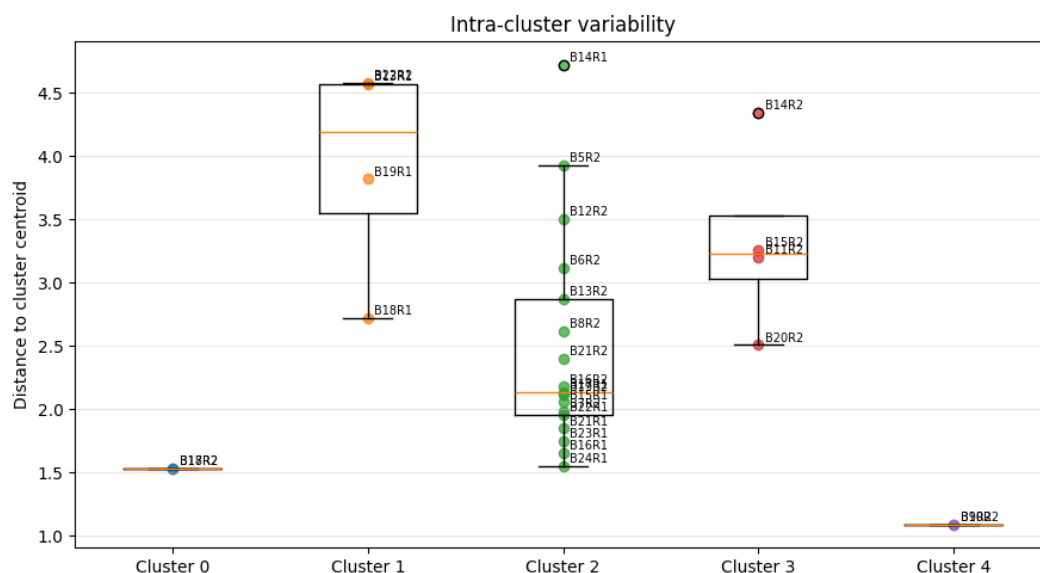


Figure 7. Distance of each cultivation from its cluster centroid. Larger values indicate greater within-cluster deviation.

Clusters 1 and 3 have relatively high mean distances (3.92 and 3.33), indicating appreciable internal heterogeneity. Cluster 2 is more compact on average (2.49) despite containing 17 experiments, although B14R1 reaches the largest distance in that cluster (4.73). Clusters 0 and 4 each contain only two members; their equal distances and near-zero distance SD are a geometric consequence of having two points around their centroid and should not be interpreted as strong evidence of biological reproducibility.

KEY FINDING

The five groups differ not only in their mean profiles but also in internal consistency. Clusters 1 and 3 are relatively heterogeneous, while the large Cluster 2 is moderately compact overall.

WHAT IT MEANS - A cluster label alone does not imply that all members behave similarly. Cultivations far from their centroid deserve direct trajectory inspection, and two-member clusters require particular caution because conventional dispersion summaries are not very informative.

8. Which features vary within each cluster?

The feature-specific within-cluster standard deviation identifies which descriptors are stable within a group and which differ substantially among its members. This complements the centroid-distance analysis by showing the source of heterogeneity rather than only its overall magnitude.



Figure 8. Feature-specific within-cluster standard deviation. Higher values indicate that members of a cluster differ more strongly for that descriptor.

The heatmap shows that heterogeneity is not uniform across features. In particular, the four-member clusters contain substantial variation in several glucose, DO, pH and biomass descriptors. Cluster 2 also contains feature-specific spread despite being the broad central group. For the two-member clusters, standard deviations should again be interpreted cautiously because they are estimated from only two observations.

KEY FINDING

Within-cluster heterogeneity is feature-dependent: some groups are coherent for certain biological descriptors but variable for others.

WHAT IT MEANS - This helps identify what to inspect next in the raw time series. A cluster may represent a shared broad phenotype while still containing meaningful differences in oxygen dynamics, pH behavior, glucose use or biomass development.

9. Interpretation, limitations and recommended next steps

This four-variable analysis reveals structured differences among the 29 cultivations and identifies several profiles that depart strongly from the main group. The results are descriptive and exploratory. They do not by themselves establish why a cultivation behaves differently. Experimental metadata and the original trajectories are required to distinguish biological effects from process settings, measurement artifacts or run-specific conditions.

Several methodological limitations should remain visible when communicating the results. First, six summary descriptors simplify the full temporal trajectories and may miss timing-specific events. Second, the correlation filter keeps one feature from a correlated pair based on column order rather than biological priority. Third, K-means assumes roughly centroid-based groups in the retained PCA space. Fourth, $k = 5$ was selected for interpretive granularity even though $k = 2$ has the highest silhouette score. Finally, clusters with only two cultivations cannot support strong conclusions about reproducibility or within-group variance.

Recommended follow-up

Priority	Action	Reason
1	Overlay the original glucose, DO, pH and biomass trajectories by cluster.	Confirm that statistical differences correspond to recognizable dynamic phenotypes.
2	Inspect high-distance members, especially B14R1 in Cluster 2 and the most distant members of Clusters 1 and 3.	Determine whether heterogeneity reflects biology, process conditions or artifacts.
3	Map clusters to experimental metadata (medium, feed, aeration, run, condition).	Test whether groups correspond to known process or biological factors.
4	Compare the $k = 2$ and $k = 5$ solutions explicitly.	Separate robust global structure from finer exploratory substructure.
5	Compare with five- and six-variable representations including Volume and off-gas CO ₂ .	Assess whether additional process/metabolic measurements add stable, nonredundant information.

KEY FINDING

The four-variable representation provides a coherent exploratory map of cultivation behavior, with glucose, DO and biomass descriptors driving much of the separation.

WHAT IT MEANS - The most useful next step is not to assign biological labels from PCA alone, but to validate the statistical groups against raw trajectories and experimental metadata, while treating the five-cluster solution as a fine-grained exploratory view.

Bottom line

The analysis identifies a broad central cultivation phenotype plus several smaller, distinctive groups. It also shows that some clusters contain substantial internal heterogeneity. Together, the inter-cluster and intra-cluster analyses provide a practical framework for selecting cultivations for closer biological and process-level investigation.